Clitics in Wolof: Syntax all the way up

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Abstract

This paper explores the properties and placement of Wackernagel-like clitics in the Niger-Congo language Wolof. I affirm the proposal initially put forth by Dunigan (1994), that Wolof clitics move in the syntax to the highest head in the extended projection of the verb, contra subsequent analyses which argue for the need for a prosodic component to Wolof clitic placement (Zribi-Hertz and Diagne 2002; Russell 2006). The main argument for the absence of a post-syntactic readjustment of clitic position comes from the behavior of the subject clitic, which is initial in the extended projection of the verb if there is no higher functional head that could attract it, but patterns with other clitics when such a head is present. I show that this is straightforwardly accounted for if the subject clitic can undergo EPP-triggered A-movement to the specifier of the highest head in the extended projection of the verb, placing it outside the c-command domain of that head. I formalize clitic movement as triggered by an EDGE-feature on the highest head in the extended projection of the verb, so that the goal feature on the clitics becomes activated in the c-command domain of the EDGE-feature, and stays inactive otherwise. Finally, I discuss clitic climbing, which reveals another property of Wolof clitics: their need to be located in the inflectional layer of the clause. Wolof clitics confirm an important property of Wackernagel-like clitics: that their movement is primarily syntactic, as argued, for example, by Ouhalla (1989) for Berber, Franks (1998/2010) for Bosnian/Croatian/Serbian, even if it in some languages involves a prosodic component. Clitic Movement of this type looks similar to other types of movement to the edges of domains. To the extent that such movements are syntactically motivated, so is the movement of Wolof-type clitics.

Keywords: *clitics, clitic climbing, infinitives, Wolof, long head movement*

1 Introduction

Since their first mention in the generative literature, *special clitics* (in the sense of Zwicky 1977) have been stubbornly resistant to analyses employing the standard array of linguistic devices both in the syntax and in the phonology. The main issues have to do with determining their positions in the clause, motivating their placement there, and determining whether they are generated in those positions or arrive there via movement. Much work on clitics proposes that they target a particular place in the clause, or that their position is tied to the position of the verb. Such an analysis is standard for Romance, which has tied clitic movement to verb movement since Kayne (1975). So-called *second position* or Wackernagel clitics, whose placement is generally independent of that of the verb, have seen all sorts of analyses over the years – syntactic, phonological, and everything in between, especially when it comes to clitics in South Slavic languages (see, e.g., Bošković 2001; Franks 1998/2010 and references therein for an overview).

Wolof clitics exhibit some familiar properties of Wackernagel clitics. Even though they do not consistently surface in the second position, Wolof clitics cluster high, immediately following overt complementizer-like elements present in every finite indicative clause, as shown in (1).[1]

(1) Wolof Wackernagel-like clitics[2]
Da=**ma=ko=fa** gis.
C=1SG.S=3SG.O=LOC see
'I saw him/her/it there.'

In non-finite clauses clitics occupy a variety of positions, depending on the presence of other clausal material. They can follow the embedded verb, as in (2a), flank the verb, as in (2b), follow the auxiliary *di*, in (2c), or cluster together at the edge of the non-finite clause, in (2d).

(2) Wolof clitics in non-finite clauses

a. Da=ma ñëw [togg=**ko**].
C=1SG.S come [cook=3SG.O]
'I came to cook it.'

b. Da=ma aaye Faatu [**mu**=togg=**ko**].
C=1SG.S prevent Fatou [3SG.S=cook=3SG.O]
'I prevented Fatou from cooking it.'

c. Da=ma ñëw [di=**ko** togg].
C=1SG.S come [IPFV=3SG.O cook]
'I came to cook it.'

d. Da=ma aaye Faatu [**mu=ko** di (>mukoy) togg].
C=1SG.S prevent Fatou [3SG.S=3SG.O IPFV cook].
'I prevented Fatou from cooking it.'

This paper has two main goals. First, I show that Wolof clitics target the highest functional head in the extended projection of the verb, as previously proposed by Dunigan (1994). Analyses along similar lines can be found in Ouhalla 1989 for Berber, and Franks 1998/2010 for (some) South Slavic clitics. Several later analyses of Wolof have challenged this, and argued for either a mixed syntax-prosody account (Zribi-Hertz and Diagne 2002; Russell 2006), or fixed clitic projections in the clausal spine where clitics are base-generated (Torrence 2013). In addition to affirming Dunigan's original proposal, I expand on the analysis of clitic climbing in Martinović 2024, arguing that clitics must establish a relationship with a head in the inflectional layer of the clause.

The second goal of this paper is to spell out the details of clitic movement. I argue that clitics have a feature that is activated and results in movement only if the clitic is in the c-command domain of its trigger, which I propose is an EDGE-feature. If the clitic is higher than the trigger, the feature remains inactive and there are no special requirements with respect to the clitic's placement. This will account for the ability of the subject clitic to be initial in examples such as (2b), and

[1] Interlinear glossing of exampes follows the Leipzig Glossing Rules, with the addition of AND = andative; CM = class marker; LOC = locative clitic; O = object clitic; OPT = optative; S = subject clitic; SG = singular; STR = strong pronoun.

[2] Unless otherwise noted, all data were collected by me in St-Louis, Senegal, and Ndombo, Senegal, during five field trips between 2014 and 2019.

explain why it behaves as other clitics when there is a higher head that attracts it, as in the finite clause in (1). I also offer a novel analysis of the optionality of clitic climbing, following the theory of movement developed in Heck 2016, 2022, 2023, which decomposes it into (i) the removal of an element into a separate workspace, and (ii) remerger of the element back into the clausal spine. Heck argues that the second step, remerger, can be delayed ("procrastinated"), which he uses to account for various apparent violations of the Minimal Link Conditions and the Strict Cycle Condition. I propose that procrastination can offer an analysis of the optionality of clitic climbing that avoids two issues that arise if the movement proceeds either in a step-wise fashion or in one fell swoop: excorporation of clitics or a violation of Relativized Minimality.

The paper proceeds as follows. In section 2 I illustrate the basic properties of Wolof clause structure and clitic placement in finite clauses, and in section 3 I present the core of the analysis. Section 4 introduces data from clitic climbing, argues that clitics are licensed in the inflectional layer of the clause, and proposes a new analysis of the optionality of clitic climbing. Section 5 investigates new data from non-finite clauses with overt subjects, and argues that they support a purely syntactic account of clitic movement in Wolof. Section 6 concludes.

2 Clitics in Wolof finite clauses

I begin this section with an overview of the distribution of pronominal elements in Wolof, and then give the necessary background on Wolof clause structure. I then give a brief overview of previous analyses of Wolof clitics, and lay out the analysis advocated for in this paper. I affirm the proposal for clitic movement made in Dunigan 1994, and resolve an empirical issue that complicated both Dunigan's and several other analyses of Wolof cliticization.

Personal and locative pronouns in Wolof have a *strong* and a *clitic* paradigm, exemplified in Table 1 and Table 2. Locative clitic forms carry a proximal/distal distinction (*-i* = proximal marker, *-a* = distal marker).[3] There is only one series of object pronouns, for both goal and theme objects (i.e., there is no distinction in Wolof between dative and accusative clitics).

	Strong	*Clitic subject*	*Clitic object*
1SG	man	ma	ma
2SG	yow	nga	la
3SG	moom	mu	ko
1PL	ñun	nu/ñu	nu/ñu
2PL	yeen	ngeen	leen
3PL	ñoom	ñu	leen

Table 1: Strong and clitic personal pronouns in Wolof

In languages with multiple series of pronouns, as noted by Cardinaletti and Starke (1999), strong and weak/clitic pronouns surface in different environments. This is also the case in Wolof,

[3] There are two forms of locative markers, one with the consonant *f-* and the other one with *c-*. I have not explored the difference between the two in their pronominal use.

Strong	*Clitic*
foofu	fa/fi
coocu	ca/ci

Table 2: Strong and clitic locative pronouns in Wolof

where only strong pronouns can be coordinated, focused, dislocated, and be complements of prepositions. ‘Weak’ object and locative pronouns behave like *special clitics* in the sense of Zwicky 1977, and almost always occupy a syntactic position different from that of a corresponding non-clitic element. Comparing strong and clitic pronouns shows that clitic pronouns are not necessarily prosodically weaker/reduced compared to the strong pronouns—2*nd* person subject clitic and 2*nd* and 3*rd* object clitics have long vowels and codas. Clitic Movement therefore cannot be attributed to the prosodically weak nature of clitics.

I show that clitics adjoin to the highest head in the extended projection of the verb; I refer to this movement as Clitic Movement. The behavior of the subject clitic, on the other hand, depends on the syntactic position that it finds itself in. If it precedes the highest head in the movement domain of clitics (here argued to be the extended projection of the verb), it surfaces in the initial position and does not necessarily cluster with other clitics (see (2b)). In that case it still forms a prosodic unit with the phonological word that follows it, but it does not undergo Clitic Movement. If the subject clitic does not precede the highest head in the extended projection of the verb, it does undergo Clitic Movement and clusters with object and locative clitics behind the highest head (as in (1)). I investigate the difference between the subject and non-subject clitics throughout the paper and argue that the positioning of the subject clitic does not require it to have a different status from other weak pronouns, contra several other analyses, and that it additionally provides evidence against a prosodic (or a mixed syntax-prosody) analysis of clitic placement.

I first establish that the subject markers in Wolof are indeed pronominal clitics, and not agreement morphology. Wolof is an SVO language with a rigid clause-internal word order and several left-peripheral positions for A′-extracted and left-dislocated elements. All finite clauses in Wolof contain a high projection which hosts complementizer-like morphemes – *sentence particles* (Dunigan 1994; Martinović 2015, 2017a, 2023; see also Torrence 2005, 2013 for a more heterogenous view of these elements). There are two types of finite clauses in Wolof (Martinović 2015, 2023), illustrated in (3): those in which a non-clitic subject is to the left of the sentence particle (examples of a *Neutral affirmative* and *Predicate focus*[4] sentence) and is obligatorily doubled by a clitic to the right of the sentence particle, and those in which an XP A′-moves in front of the particle, and the subject remains below it (examples of an *Exhaustive Identification* sentence and a *Relative clause*). The sentence particles are boxed.

(3) Wolof clause types

a. Xale yi lekk-**na**=ñu gato bi.
child the.PL eat-C=3PL.S cake the.SG
“The children ate the cake.” Neutral affirmative

[4] Labels referencing information-structural properties of various clauses are here used purely descriptively.

b. Xale yi $\boxed{\textbf{da}}$=ñu lekk gato bi.
child the.PL C=3PL.S eat cake the.SG
"The children ATE the cake." Predicate focus

c. Gato bi $\boxed{\textbf{la}}$ xale yi lekk.
cake the.SG C child the.PL eat
"It's the cake that the children ate." Exhaustive Identification

d. gato b-$\boxed{\textbf{u}}$ xale yi lekk
cake CM-C child the.PL eat
'a cake that the children ate' Relative clause

I follow Dunigan (1994) and Martinović (2015, 2017b, 2023), who argue that sentence particles all occupy the same position in the clause. They are in complementary distribution with one another, they appear to carry clause-typing information, and a sentence particle is obligatory in order for the clause to have tense and negation (Njie 1978, 1982). Elements to the left of sentence particles behave as if they were in the left periphery. I therefore identify this position as a (low) C head.[5]

There are a number of syntactic differences between the two clause types in (3). Most strikingly, in clauses which do not involve A′-extraction, as (3a) and (3b) above, a non-clitic subject can only be to the left of the sentence particle (*xale yi* 'the children') and is obligatorily doubled by a clause-internal clitic (*ñu*). The non-clitic subject cannot occur to the right of the sentence particle, regardless of the presence or absence of subject marking, indicated with brackets around the subject clitic in (4). This makes it appear as if the element encoding the φ-features of the subject is agreement, and the reason that the non-clitic subject is banned from a post-agreement position is due to the fact that it needs to move to the specifier of Agr, for example.

(4) Only one subject below C

a. *Lekk-$\boxed{\text{na}}$(=ñu) xale yi gato bi.
eat-C=3PL.S child the.PL cake the.SG
intended: *'The children ate the cake.'*

b. *$\boxed{\text{Da}}$(=ñu) xale yi lekk gato bi.
C=3PL.S child the.PL eat cake the.SG
intended: *'The children ATE the cake.'*

In A′-extraction of a non-subject, however, the clause-internal subject can either be a clitic, as in (5), or a non-pronominal DP, as in (6). The clitic and non-clitic subject cannot co-occur to the right of the sentence particle, but one is obligatory. In local A′-extraction of a subject pronoun, the pronoun is found to the left of the sentence particle and a clause-internal subject clitic is not possible (for discussion of subject pronoun extraction see p.9, example (16a)).

(5) Clitic subject in A′-extraction

[5] Wolf also has a high embedding complementizer, *ni*, which can embed all clauses with sentence particles. In a more articulated left periphery à la Rizzi (1997), *ni* would be in Force, and C would be closest to Fin, which is how Zribi-Hertz and Diagne (2002) and Torrence (2003) label the particle *na*. Given that *na* does not occur in all finite clauses, and is incompatible with particles that occur in A′-movement (subject and non-subject focus and relative clauses), which are traditionally considered to be complementizers, I label all these heads uniformly as C.

a. Gato bi $\boxed{\text{la}}$=ñu lekk.
cake the.SG C=3SG.S eat
'It's the cake that they ate.'

b. gato b-$\boxed{\text{u}}$=ñu lekk
cake CM-C=PL.S eat
'a cake that they ate'

(6) Non-clitic subject in A′-extraction

a. Gato bi $\boxed{\text{la}}$(=*ñu) xale yi lekk.
cake the.SG C(=*3SG.S) child the.PL eat
'It's the cake that the children ate.'

b. gato b-$\boxed{\text{u}}$(=*ñu) xale yi lekk
cake CM-C(=*PL.S) child the.PL eat
'a cake that the children ate'

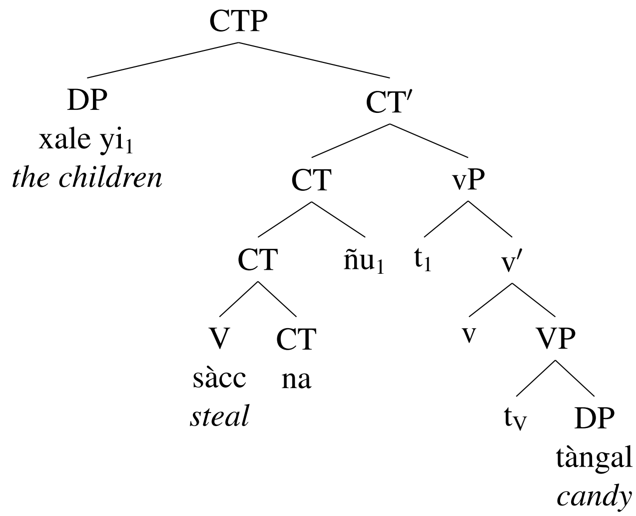
The comparison between the two clause-types shows that the subject clitic to the right of C should not be treated as agreement (contra Torrence 2005, 2013; Zribi-Hertz and Diagne 2002). In Martinović 2015, 2023, I propose that clauses as in (3a)-(3b) involve the dislocation of the non-clitic subject to the left periphery, with the clitic being the true clause-internal subject.[6] Specifically, I argue that, in clauses which obligatorily dislocate subjects, the clause-internal subject position is not available because the features of C and T are bundled into one head, therefore only making one specifier position available. In A′-movement constructions, on the other hand, C and T head separate projections, therefore making one specifier position available for the subject, and another for the A′-moved element. The structure of the two clause types is shown in (7)-(8), and I assume this in the rest of the paper. The bundling of the C and T head will be relevant for cliticization in infinitives with overt subjects. (I assume the verb respects the Head Movement Constraint in moving to CT or T, but omit this from the phrase markers for simplicity.)

(7) Bundled CT

a. Xale yi sàcc na=ñu tàngal.
child the.PL steal C=3PL.S candy
'The children stole candy.'

[6] It has been argued for a variety of Niger-Congo languages, mostly in the Bantu family, that pre-verbal subjects in finite indicative clauses are topical/in some left-peripheral position (a.o., Henderson 2006; Schneider-Zioga 2000, 2007; Pietraszko 2021).

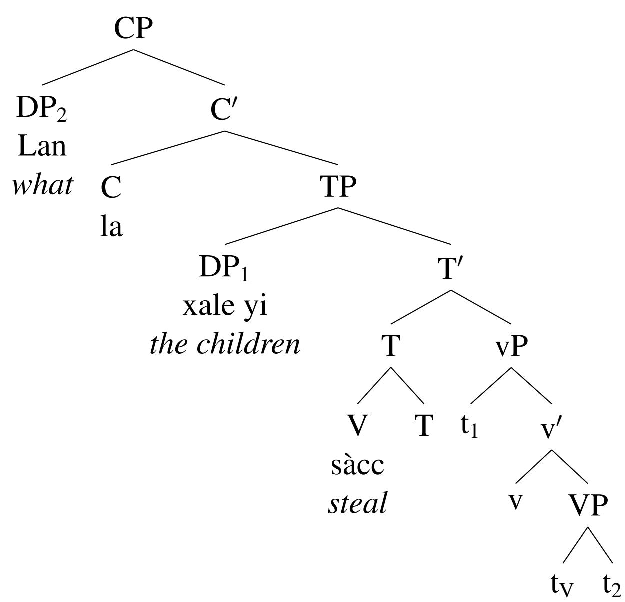
b.



(8) Split C and T

a. Lan la xale yi sàcc?
what C child the.PL steal
'What did the children steal?'

b.



In finite matrix clauses, clitics target the position to the immediate right of the sentence particle, clustering in the order SUBJ≺OBJ≺LOC. The examples in (9) show clauses with no A′-movement and non-clitic subjects to the left of C (a-c), and clauses with A′-movement (d-e).[7]

(9) Clitic position in finite clauses

[7]Torrence (2005, 2013) argues that conditional clauses as in (9e), and temporal clauses, are a type of a relative clause, and therefore involve movement of a null operator to Spec,CP. The details are not relevant to our purposes, what is important is that *su* is a C head.

a. (Xale yi) lekk-$\boxed{\textbf{na}}$=<u>ñu=ko=fi</u>.
child the.PL eat-C=3PL.S=3SG.O=LOC
"The children ate the cake."

b. (Xale yi) lekk-ul-$\boxed{\emptyset}$=<u>ñu=ko=fi</u>.
child the.PL eat-NEG-C=3PL.S=3SG.O=LOC
"The children didn't eat the cake."

c. (Xale yi) $\boxed{\textbf{da}}$=<u>ñu=ko=fi</u> lekk.
child the.PL C=3PL.S=3SG.O=LOC eat
"It's that the children ate the cake."

d. Demb $\boxed{\textbf{la}}$=<u>ñu=ko=fi</u> lekk.
yesterday C=3PL.S=3SG.O=LOC eat
"It's yesterday that they ate it."

e. $\boxed{\textbf{su}}$=<u>ñu=ko=fi</u> lekk-oon, ...
C=3PL.S=3SG.O=LOC eat-PST
'If they had eaten it here,...'

The non-clitic subject to the left of C in (9a)-(9c) is not focused, and it does not have to be a topic—sentences such as (9a)-(9b) can be uttered out-of-the-blue (Russell 2006; see Martinović 2015, 2023 for a detailed analysis of the positions of subjects in the left periphery). Any clause can have a topic; if the subject to the left of C in clauses such as (9a)-(9d) is topicalized, it is followed by a pause (Rialland and Robert 2001). The optional non-topic subject in (9a)-(9c) forms an intonational unit with the material that follows it. The clitics therefore do not occur in a consistent position from the left edge of the intonational phrase: they can follow the second phonological word (the subject DP and the verb with its affixes and the complementizer in (9a)-(9c)), or the first phonological word if the subject DP to the left of C in those examples is absent; the same variation in clitic placement is found A′-movement constructions in (9d)-(9e). Clitics cannot be initial in finite clauses, which follows straightforwardly from an analysis in which they target the highest functional head in the extended projection of the verb, the sentence particles, which in Wolof I identify as C heads.

The position of the clitics is higher than the structural subject position (Russell 2006), which can be seen when the clause-internal subject is not pronominal (in A′-movement clauses, where this is possible). Compare (10a), which has a clitic subject, and (10b), with a non-clitic subject.

(10) Clitics are higher than structural subjects

a. Yow $\boxed{\textbf{la}}$=<u>ñu=ko=fa</u> wax.
you C=3SG.S=3SG.O=LOC say
'It's you that they said it to there.'

b. Yow $\boxed{\textbf{la}}$=<u>ko=fa</u> **jigéen yi** wax.
you C=3SG.O=LOC woman the.PL say
'It's you that the women said it to there.'

The examples above show that clitics always occupy the same position when a sentence particle is present, independently of the presence or position of any other element in the clause (the verb, the non-clitic subject, and A′-moved XP).

Clitics also occur in structures which do not contain overt C elements, in which case they do not necessarily cluster together. For example, there are clauses that occur in running narrative contexts (story-telling) or in proverbs. They are unembedded, but they cannot contain any inflectional morphology (tense, aspect, negation); they must acquire temporal reference from context, or be directly preceded by a temporal adverbial phrase. I refer to them as *minimal clauses* (Zribi-Hertz and Diagne 2002); an example is given in (11). Pronominal clitics can occur in such structures, as shown in (12).

(11) Minimal clause
Xale yi lekk gato bi ci wañ bi.
child the.PL eat cake the.PL in kitchen the.SG
'The children eat/ate/will eat/... the cake in the kitchen.'

(12) Clitics inside minimal clauses
Ñu=lekk=**ko**=**fa**.
3PL.S=eat=3SG.O=LOC
'They ate/eat/will eat it there.'

The fact that the subject clitic can be clause-initial shows that Wolof does not have a prosodic ban against initial clitics; compare, for example, with Warlpiri, which Legate (2008) argues involves syntactic clitic placement, except in one instance where postsyntactic re-ordering is triggered due to the clitics being enclitics and therefore needing to lean onto another element to their left. No such requirement appears to be placed on Wolof clitics. This example additionally shows that clitics do not occur in a consistant position from the left edge of the intonational phrase.

As expected, Wolof clitics form a phonological word with the material that they cliticize to. This can be seen from the application of vowel harmony to the clitic cluster, shown in (13). Wolof has ATR-harmony, which spreads from roots to affixes and clitics. Note that vowel harmony spreads only inside phonological words, and each phonological word contains only one root; for example, the non-clitic subjects to the left of C in (9) do not harmonize with C and clitics. In the remainder of the paper, I do not represent vowel harmony but consistently use the -ATR spelling for clitics and affixes.

(13) Vowel harmony in clitics
a. Lekk-na=∅=leen=fa.
eat-C=3SG.S=3PL.O=LOC
'He ate them there.' -ATR verb, -ATR clitics
b. *Lekk-në=∅=léen=fë.
eat-C=1SG.S=3PL.O=LOC
intended: *'He ate them there.'* -ATR verb, +ATR clitics
c. Dóor-në=∅=léen=fë.
hit-C=3SG.S=3PL.O=LOC
'He hit them there.' +ATR verb, +ATR clitics
d. *Dóor-na=∅=leen=fa.
hit-C=3SG.S=3PL.O=LOC
intended: *'He hit them there.'* +ATR verb, -ATR clitics

Additionally, subject clitics have a different form depending on the sentence particle they are adjacent to. For example, the 1st person subject clitic surfaces as *a* in Neutral affirmative clauses (with the sentence particle *na*), but as *ma* if negation is on the verb, in which case the sentence particle is phonologically null, shown in (14). The subject clitics and C form portmanteau morphemes in 2nd person singular and plural.

(14) Subject clitic varies in form

a. Lekk-$\boxed{\text{na}}$=**a** ceeb.
eat-C=1SG.S rice
'I ate rice.'

b. Lekk-u(l)-$\boxed{\emptyset}$=**ma** ceeb.
eat-NEG-C=1SG.S rice
'I didn't eat rice.'

(15) 2nd person + C portmanteau

a. Lekk=$\boxed{\textbf{nga}}$ ceeb.
eat=C.2SG.S rice
'You ate rice.'

b. Lekk=$\boxed{\textbf{ngeen}}$ ceeb.
eat=C.2PL.S rice
'You(pl) ate rice.'

This kind of variation in form is present in most persons, but not all – for example, the 1st and 3rd person plural pronouns remain the same in all environments. The phonological form of object and locative clitics does not change depending on the morphosyntactic context. I take this to mean that subject clitics incorporate into C, whereas other clitics do not. This is not relevant for the discussion of Clitic Movement.

Finally, we need to resolve a puzzle related to the surface form of subject pronouns that move to Spec,CP, which has complicated some previous analyses of clitics, discussed in the following subsection. Pronouns surface in their strong form in left-peripheral positions, if they are focused, as complements of prepositions, and in coordination. At first glance, Wolof appears to be a counterexample to the generalization that pronouns always surface as strong in focus/left-peripheral positions. An object pronoun that A′-moves to Spec,CP can only surface in its strong form, as expected, shown in (16a), not as a clitic, in (16b). In case of subject extraction however, it appears that the weak version of the pronoun, in (17a), surfaces in Spec,CP, and not the strong one, as in (17b).[8]

(16) A′-extraction of object pronoun

a. Ñoom la Usmaan gis.
3PL.STR C Oussman see
"It's them who Oussman saw."

b. *Leen la Usmaan gis.
3PL.O C Oussman see

(17) A′-extraction of subject pronoun

a. Ñu-a (>ñoo) gis Usmaan.
3PL.S-C see Oussman.
"It's them who saw Oussman."

b. *Ñoom-a gis Usmaan.
3PL.STR-C see Oussman.

The asymmetry between subject and object pronouns in extraction is, however, only apparent. They key clue comes from 2nd person singular and plural, where there is no phonological similarity between weak and strong pronouns (see Table 1). In case of 2nd person subject extraction to Spec,CP, the pronoun does not surface in its weak form, as *nga/ngeen*, but as a truncated strong form, *ya* (full strong form is *yow*), in the singular, and as the full strong pronoun *yeen* in the plural.

(18) 2nd person subject pronouns in Spec,CP

[8] The A′-movement complementizer in Wolof exhibits a subject/non-subject asymmetry, surfacing as *a* in subject extraction and as *la* in extraction of any other constituent.

a. Ya-a gis Usmaan.
2SG.STR-C see Oussman
"It's you(sg) who saw Oussman."

b. *Nga-a gis Usmaan.
2SG.S-C see Oussman

c. Yeen-a gis Usmaan.
2PL.STR-C see Oussman
"It's you(pl) who saw Oussman."

d. *Ngeen-a gis Usmaan.
2PL.S-C see Oussman

The generalization that Spec,CP is a position in which only strong pronouns can surface therefore holds for Wolof as well, and subject pronouns in Spec,CP are truncated versions of the strong forms. This is important, as the assumption that subject pronouns in Spec,CP are clitics has introduced complications into previous analyses of cliticization in Wolof, as I discuss in §2.1.

Another argument in favor of this analysis are pronouns in fragment answers. Fragments have been argued to have full sentential structures, to account for their propositional character. Merchant (2004) proposes that the fragment moves to the specifier of a left-peripheral head, with the TP elided, and fragment answers in Wolof, which can be followed by the A′-movement complementizer *(l)a*, support this. Only strong pronouns can be fragment answers, as shown in (20), regardless of whether or not they are followed by the complementizer *(l)a*, and regardless of their grammatical relation (Martinović 2018). The fragments in (20) can both be answers to either of the questions in (19).

(19) Subject/object questions

a. Kan a gis Aali?
who C see Ali
'Who saw Ali?'

b. Kan la Aali gis?
who C Ali see
'Who did Ali see?'

(20) Subject/object fragments

a. Man/*Ma.
1SG.STR/1SG.S
'Me'.

b. {Man/*Ma} (l)a.
1SG.STR/*1SG.S C
'Me.'

Fragment answers show that the form of the pronoun that surfaces in Spec,CP is indeed the strong form, which becomes reduced in subject extraction when the CP-layer is followed by overt material. This truncation seems to be a phonological phenomenon, and it occurs not only in Spec,CP, but also in coordination, where the subject pronoun that is the first conjunct ends up being truncated in the same way as subject pronouns in Spec,CP. I offer no analysis of the truncation of strong pronouns, as it is not relevant for cliticization.

It is here important to note that the overt subject pronoun in non-finite clauses is *not* a truncated strong pronoun, as the 2*nd* person pronoun surfaces in the clitic form *nga*, and not the reduced strong pronoun form *ya*. (21) shows that subject pronouns in initial positions of minimal clauses are indeed clitics.

(21) Subject pronouns in minimal clauses are clitics

a. Nga=gis=ko.
2SG.S=see=3SG.O
'You saw/see/will see it.'

b. *Ya=gis=ko.
2SG.STR=see=3SG.O

The discussion thus far has highlighted two important properties of Wolof clitics: (i) they do not surface in a consistent position with respect to the left edge of the clause and/or an intonational phrase, and (ii) a clitic may be initial in the intonational phrase. I shall argue that these properties can be captured by a purely syntactic account of clitic placement. Before going into details of clitic movement in Wolof, I briefly summarize previous analyses of Wolof cliticization.

2.1 Previous analyses

This paper affirms the analysis originally proposed by Dunigan (1994), who argues that clitics move as heads to the highest functional head in the extended projection of the verb in the sense of Grimshaw 1991, where the extended projection is composed of a lexical head and its projection plus all of the functional heads that represent features of the lexical head, and their projections. Dunigan proposes that clitics move via Long Head Movement, similarly to what I propose in (25) below, though because she treats subjects in Spec,CP of A′-movement constructions as clitics, and not truncated strong pronouns, she is forced to treat subject clitics differently, and proposes that they move as phrases, with some additional assumptions needed to get them to precede object and locative clitics when they follow C. I have shown above that this view of subject pronouns in Spec,CP is mistaken, but I will argue that subject clitics can undergo A-movement.

Zribi-Hertz and Diagne (2002) (ZHD) criticize Dunigan's purely syntactic account, and offer a mixed syntax/prosody analysis. The issue that ZHD take with Dunigan's analysis has to do mostly with their disagreement of her treatment of subject pronouns as clitics. They choose to treat them as agreement in clauses which I analyze as having non-clitic subjects to the left of C and only clitics as clause internal subjects (i.e., subjects below C); see (3) and the discussion that follows. Treating subject clitics as agreement in only these cases forces ZHD to divorce the behavior of subjects from other clitics. In all other types of clauses subject clitics cannot be agreement, as they are in complementary distribution with non-pronominal subjects, as shown again in the A′-movement example in (22):

(22) Only one subject below C
Gato la {xale yi}/{ñu} lekk.
cake C child the.PL/3PL.S eat
'It's cake that the children/they ate.'

ZHD take this to mean that only some finite clauses exhibit subject-verb agreement, whereas other ones do not. I argue extensively against this view in Martinović 2015, 2023. In a broader Niger-Congo context, it has been independently shown for a variety of languages that subjects in finite declarative clauses are in some left-peripheral/dislocated position, as opposed to subjects in other kinds of clauses (notably, A′-movement constructions); Wolof is therefore not unusual in obligatorily dislocating the subject in certain kinds of clauses. Given that dislocated elements in Wolof must be co-indexed with a clause-internal pronoun, the occurrence of the clitic is expected. ZHD are also troubled by the apparent occurrence of clitics in Spec,CP in A′-movement constructions, which also forced Dunigan to treat subject clitics differently from other clitics; we can dispense with this issue as well, as I have shown that pronominals in Spec,CP are truncated strong pronouns, and not clitics. Ultimately, the definition of cliticization that ZHD offer is almost identical to Dunigan's, but with an added prosodic component:

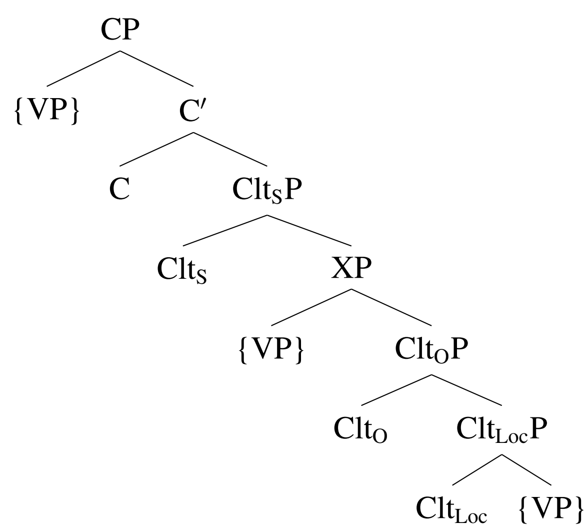
(23) ZHD definition of clitic placement (p.877)
Attach object and locative clitics to the prosodic word which contains the topmost head of their extended V domain.

The prosodic component is needed precisely because ZHD sometimes treat subject clitics as agreement in a separate head, which first must incorporate in the highest head (in my analysis C), and they argue that this incorporation is postsyntactic. Since other clitics do not precede, but follow the subject clitic, their placement must now also be postsyntactic. However, we can see from ZHD's definition that the actual clitic placement is syntactic, as it references a syntactic position. The PF then needs to know what the "topmost head" in the extended domain of the verb is, even though the structure has already been linearized and prosodic words have been formed. Ultimately, I believe that it is exactly clitics that show that ZHD's view of Wolof clausal structure is not correct.

Another analysis invoking prosodic readjustment is that of Russell (2006), who proposes that all clitics in finite clauses move to Spec,TP, and are then moved as a block to C postsyntactically. Clitics in infinitival clauses disprove this analysis, so I will comment on it in more detail at the end of section 5.

Finally, Torrence (2013), along the lines of Sportiche (1996), assumes that clitics occupy fixed projections in the clause, as in (24), and that the differences in their surface position with respect to other elements in the clause is the result of how high the verb or the verb phrase moves.

(24)



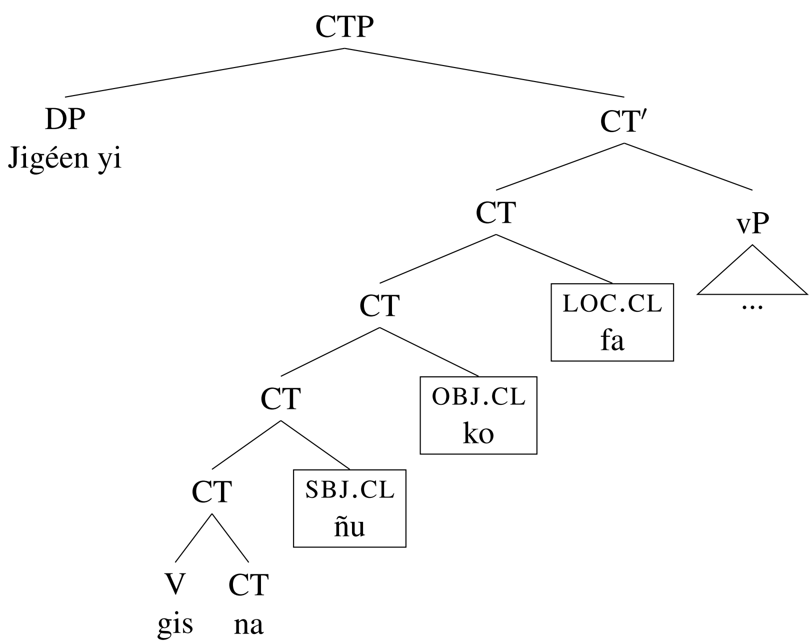
Torrence accounts for the different clitic positions via a combination of verb movement, remnant VP movement, and, in some cases, independent head movement of the negation suffix. The different types of movement do not appear to me to be motivated or constrainable in any way. For example, Torrence (2003) assumes that the verb moves as a head in clauses in which negation is not present, but that the whole VP moves if it is, and additionally, just in that case, negation must head-move as well so as to get the correct morpheme ordering. (This is employed to account for the behavior of the optional past tense morpheme *oon* and not directly relevant here, but see Martinović 2019 for an alternative analysis.) There is nothing principled that blocks verb movement and forces VP movement in some structures but not in others. It is also not clear exactly where the different clitic positions sit in the clausal spine, and what the position for VP movement (labeled XP in (24)) would be.

My goal in this paper is to affirm Dunigan's initial proposal, and to additionally show that a uniform analysis can be given for both subject, and object/locative clitics. Additionally, we shall see that clitics can be a useful tool for identifying the amount of functional structure in a clause. I represent the moved clitics as in (25), right-adjoined to the head that they target.

(25) Clitics adjoin to the right of the highest head

a. Jigéen yi gis na=ñu=ko=fa.
woman the.PL see C=3PL.S=3SG.O=LOC
'The women saw her/him/it there.'

b.



In the next section, I lay out the details of the analysis of Clitic Movement in finite clauses.

3 Clitic movement

Different triggers have been proposed for clitic movement. It has been argued that clitics are defective elements, not just prosodically, but in every domain (Sadock 1991), and this deficiency has often been given as an argument for their special syntactic behavior (e.g., Cardinaletti and Starke 1999; Déchaine and Wiltschko 2002; Franks 2012). In such accounts the movement of clitics is triggered by their need to be licensed in some way, so they are generated in or move to agreement projections, or to a position where they can be case-licensed. Additionally, various prosodic requirements may be placed on clitics, which may cause them to occupy specific positions in a prosodic word; this appears to be the case with second position clitics.

The analysis here is similar to the one in Franks (1998/2010). While arguing that BCS clitics do move to appropriate Agr positions for case-checking purposes, Franks notes that they show up as high in the tree as possible, specifically, in the highest head position. He proposes a variation of Wackernagel's traditional insight connecting the V2 phenomenon with second position clitics, in the sense that whatever triggers verb movement is also responsible for clitic movement (see also Anderson 1993, 1995, 2005). Given that clitics in BCS are not always next to the verb, Franks proposes that all languages are V2 at LF and that the verb always undergoes head movement, but in some cases does not reach the highest position overtly. Clitics move because they are "looking for their verbs", "knowing" that the verb must be high. Since clitics do not actually know where the verb ends up being pronounced overtly, they just move as high as they can.

Saying that clitics are moving to a higher position because verbs are also moving to a higher position is simply restating the observation that some elements gravitate towards the clausal edge.[9] Another type of element that tends to move very high are of course *wh*-words. We accept that certain elements move to higher positions in the clause, and we have developed formal machinery to capture the relationship between the trigger of movement and the moving element, but we seem to be less concerned with the *why* when it comes to, for example, verb movement or *wh*-movement. Verb movement has been tied to, for example, the high position of Tense (e.g., Den Besten 1989), or inflectional morphology in general, but in a language like Wolof, which for the most part behaves like a tenseless language (Bochnak and Martinović 2018; Bochnak and Martinović 2019), and where there is barely any inflectional morphology to speak of and the verb still moves fairly high, the trigger must be something else. *Wh*-movement has been linked to the high position of focus in the clausal architecture (e.g., Rudin 1988), but focused elements do not always move in languages in which *wh*-words move. I am here therefore not concerned with "why" clitics move, in that I do not consider it to be the right question, or even a relevant one.

I propose that the movement of clitics is just another movement triggered by the domain edge, and I formalize this through an EDGE-feature ([EF]) on the highest head. It is important to reiterate that Wolof clitics do not move to the edge of the intonational phrase, as the head that they attach to may or may not be preceded by a separate phonological word, without affecting clitic placement. Edge features have been proposed in the context of cyclic A′-movement, in order to allow the wh-word to move to the edge of a phase (i.e., spell-out domain) and thus escape transfer, remaining visible to the higher probe (e.g., Chomsky 2000, 2001; Kandybowicz 2009; Müller 2011; Korsah and Murphy 2020). Various movements to the edge differ in non-trivial ways: in case of V2 (to the extent that that one such movement), only one verb, the highest one, moves to C, whereas in the case of *wh*-movement some languages move only one, and others move multiple phrases to the edge. Clitic movement appears more similar to head movement than phrasal movement. Clitics do not move cyclically like wh-words, and neither do verbs. Additionally, wh-movement in Wolof is possible only out in clauses with a wh-movement complementizer, whereas clitics move to the edge of any clause, regardless of the kind of C it contains. If these movements to the edge are caused by one and the same trigger, something else will need to account for the differences between them. I leave this aside as a larger puzzle.

The next question has to do with how exactly the probe-goal relationship between the trigger of

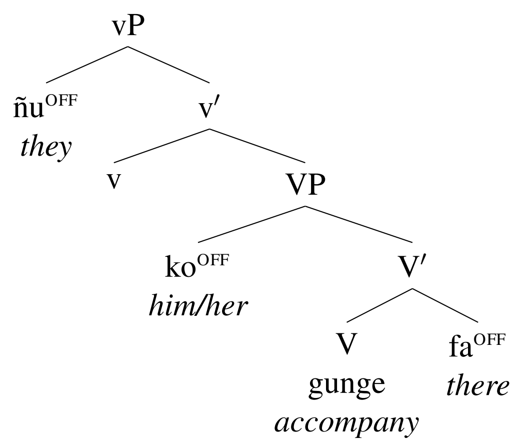
[9]On arguments agains unifying clitic movement with V2, and against a unified trigger for all verb movement to C, see e.g. Migdalski 2010.

movement and the clitic can be captured. We saw in (12) that the subject clitic is sometimes initial, making it seem like it does not undergo Clitic Movement. Any description of clitic placement that requires them to be in a particular position (e.g., *Clitics follow the highest head in the extended projection of the verb*) therefore cannot be entirely correct. I propose that the relevant feature on the clitics that is attracted by [EF] is akin to a switch, which has an ON and an OFF position. When the derivation is sent to PF, the switch has to be in the OFF position. It gets switched ON if the pronoun is in the c-command domain of [EF], and switched OFF by adjoining to the head carrying [EF]. If a clitic is not in the c-command domain of [EF] (because it moves to the specifier of the highest head for EPP, as I argue in section 5), the feature never gets switched ON and the clitic does not participate in Clitic Movement. Note that [EF] is not an unvalued feature that must be checked; it is simply the property of the highest head in the extended projection of the verb to which clitics are sensitive. This conceptualization of the relationship between trigger of clitic movement and the 'feature' on the clitic which is sensitive to the trigger will allow us to maintain a uniform analysis for all clitics, instead of introducing an exemption for the subject clitic in non-finite clauses, which, I shall argue, is not supported by empirical evidence. In this section, I limit the discussion to clitic movement in finite clauses; cliticization in infinitives is discussed in the following two sections.

To illustrate the mechanism, consider the example in (26a) with the ditransitive verb *gunge* 'accompany', which takes a theme and a locative argument. For simplicity, I place them in the specifier and complement positions of V. The tree in (26b) shows the clitics in their base-generated positions when the vP has been built, which I assume are the same as those where their non-clitic counterparts are generated.

(26) a. Gunge na=ñu=ko=fa.
accompany C=3SG.S=3SG.O=LOC
'They accompanied him/her there.'

b.



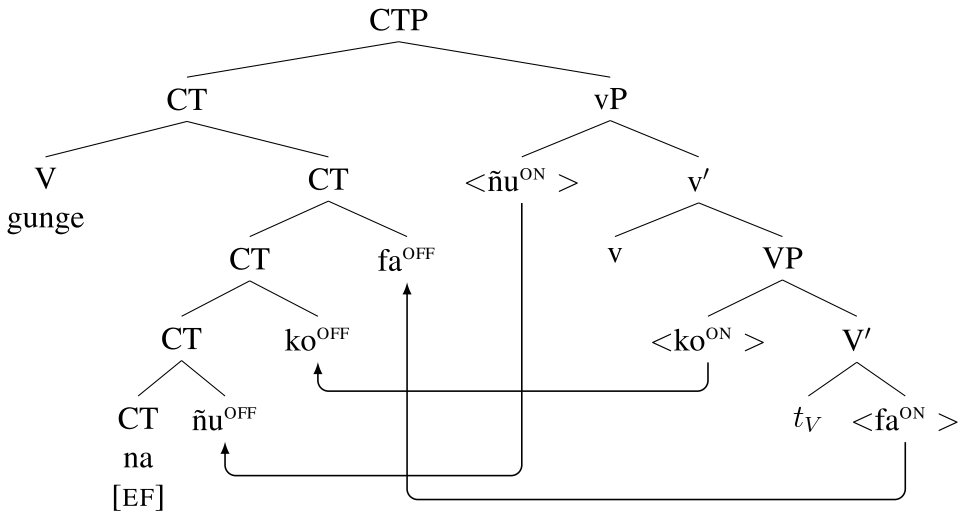
Recall that I assume that the head that hosts the complementizer (C) and the head that hosts the subject (T) in its specifier are bundled into one head in structures as in (26a)—CT. (28) represents the tree in which the CT head has been merged (any other heads that may occur between the vP and CT, such as aspect, or negation, are ignored here). This is the highest head in the extended projection of the verb, and carries [EF], which flips the switch of the clitics it c-commands in the ON position, and that, in turn, triggers their movement to CT. Clitics cluster in the order SUBJ≺OBJ≺LOC, and object clitics are additionally ordered amongst themselves along two dimensions:

(27) Clitic ordering

a. 1st person ≺ 2nd person ≺ 3rd person
b. long vowel ≺ short vowel

Due to the fact that there is a non-syntactic component to clitic ordering, and that their final order is not necessarily derived syntactically, I do not discuss clitic ordering here. When a clitic adjoins to the head bearing the [EF] feature, its switch is again set into the OFF position. (I do not represent the intermediate movement positions of the verb for simplicity.)

(28)



I do not assume that clitics move through the edge of vP, as I have no evidence that the vP is a phase in Wolof; in other words, I propose that it is only the highest head in the extended projection of the verb that carries the relevant [EF]. This is compatible with the view of phasehood in Bošković 2014, where only the highest phrase in the extended projection of a lexical category functions as a phase.

As an aside, note that there is no EPP-triggered movement of the subject pronoun here. The specifier of CT may be occupied by a non-clitic subject (Martinović 2015, 2023), obligatorily doubled by a clitic, which is in these constructions the only licit clause-internal argument. The details are not relevant here; see the original work for extensive arguments for these claims. What is relevant is that there is no evidence for another subject position below CT, so we can assume that the subject clitic moves directly from Spec,vP to CT.

We now turn to A′-constructions, where I will argue that the subject clitic does participates in EPP-triggered A-movement. Recall that I assume that, unlike non-extraction clauses, A′-movement constructions have split C and T heads, and Spec,TP is available as a clause-internal derived subject position (see (8).) Consider first (29a), where an adjunct is A′-moved to Spec,CP, and the clause-internal subject is not a clitic. Note that the non-clitic subject is below the position of clitics (shown by the locative clitic *fa*). The subject is also not in-situ in Spec,vP, as it is higher than a verb suffixed with the optional tense morpheme *oon* (Bochnak and Martinović 2019). Both the subject and the verb are above low adverbial phrases, which are found between the verb and the object (Martinović 2015, 2023). I therefore take the subject to be in Spec,TP where it, by hypothesis, moves due to the EPP. In (29b), the subject is a clitic, and it immediately follows C, preceding the locative clitic.

Given that subjects in Wolof do not remain in situ, but move to a higher position, it stands to reason that the clitic subject also moves to Spec,TP before cliticizing to C. Additional evidence for this claim is given in the following section, where we consider clitics in non-finite clauses.

(29) The positions of non-clitic and clitic subject in non-subject extraction

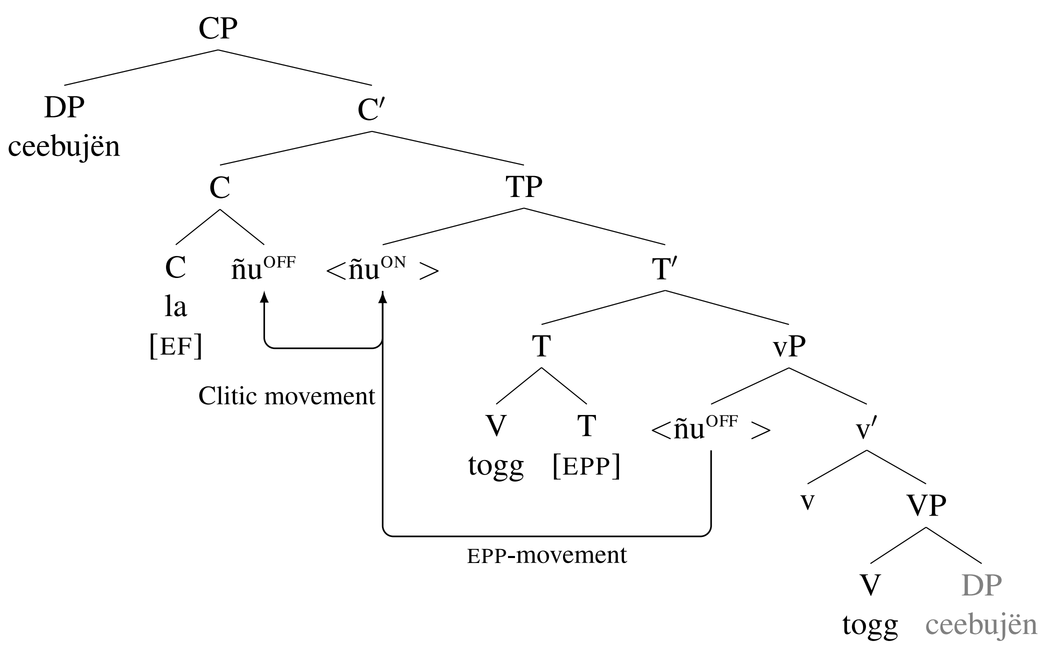
a. Demb la=fa **samay** **xarit** togg-oon bu gaaw ceebujën.
yesterday C=LOC 1POSS.PL friend cook-PST C_{REL} be.quick ceebujen
'It's yesterday that my friends quickly cooked ceebujen there.'

b. Demb la=**ñu**=fa togg-oon bu gaaw ceebujën.
yesterday C=3PL.S=LOC cook-PST C_{REL} be.quick ceebujen
'It's yesterday that they quickly cooked ceebujen there.'

The derivation involving EPP-movement of the subject clitic followed by adjunction to C is illustrated for (30a) in (30b). When T is merged, the subject clitic moves to Spec,TP to check EPP. Subsequently, C is merged, and triggers (string vacuous) Clitic Movement.

(30) a. Ceebujën la=ñu togg.
ceebujen C=3PL.S cook
'It's ceebujen that they cooked.'

b.



Clitics commonly show mixed behavior, in that they appear to have properties of both heads and phrases/minimal and maximal projections (e.g., Sportiche 1996). I adopt this position here, and assume that the subject clitic moves to Spec,TP as a phrase, but adjoins to C as a head. That clitics can undergo different kinds of movements, either phrasal movement or (long) head movement, remains a stipulation in my analysis. The different types of movement are currently tied to the kind of feature that is triggering the movement: EPP triggers phrasal movement, while the EDGE-feature triggers long head movement. This difference would ideally be derived, which I do not do here. I hope to show in the remainder of the paper that relegating Clitic Movement to postsyntax has no empirical support, on the contrary, that it makes wrong predictions about the behavior of clitics in infinitives with overt subjects. I therefore argue that both A-movement of the clitic and Clitic Movement occur in the syntax, and leave the details of how to derive their distribution for future work.

In this section I introduced the mechanism of Clitic Movement, which involves a novel conceptualization of the relationship between the movement trigger and the moved element. I have proposed that clitics are attracted by the EDGE-feature in case they are in its c-command domain. Since Wolof finite indicative clauses always contain a C-head, [EF] always c-commands all clitics, and therefore they all always move and cluster to the right of C. In section 5 I discuss infinitival complements with overt subjects, where the subject clitic occupies the specifier of the highest head in the extended projection of the verb, and the lower clitics follow it. Treating Clitic Movement as occurring only if the clitic is in the c-command domain of the trigger of movement allows us to capture the dual behavior of the subject clitic.

The following section expands our investigation of clitics to non-finite clauses in subject control constructions, which reveal an additional property of clitics—that they must occupy the inflectional layer of the clause.

4 Clitics and the inflectional layer

In this section, I discuss cliticization in control constructions, focusing on patterns of clitic climbing. I present evidence from Martinović 2024, showing that clitics have a feature that needs to be checked by a head in the inflectional layer of the clause, so that the absence or presence of the inflectional layer in the infinitival complement makes clitic climbing either obligatory or optional. I also argue that the movement of the clitic to the edge and the checking of its inflectional feature are independent of one another. Finally, I develop a novel analysis of the optionality of clitic climbing, relying on the theory of movement by Heck (2016, 2022, 2023), who proposes that it consists of two steps: the removal of an element from the derivation into a workspace, and subsequent remerger back into the derivation. Heck proposes that the second step can be delayed (*procrastinated*), and I show how this allows us to straightforwardly derive the apparent optionality of clitic climbing.

4.1 Non-finite complements and clitic climbing

The structure and size of infinitives is explored in detail in Martinović 2024, so I will keep the discussion here brief. Wolof non-finite complement clauses are characterized by the following properties: none of them contain an overt sentence particle/complementizer, they cannot contain negation, and overt subject pronouns are obligatory whenever the matrix verb takes a DP object or when the embedded subject is not controlled, but prohibited otherwise. The verb in non-finite clauses is in its bare form, i.e., there is no special infinitival morphology. In this subsection we are concerned with clauses as in (31), which have the properties of typical subject control constructions: the embedded subject is unpronounced, control is obligatory, and only sloppy reading is available under ellipsis.

(31) Non-finite complement clause

Jéem na=ñu [may miskin yi xaalis].
try C=1PL.S [give poor the.PL money]
'We tried giving money to the poor.'

In Martinović 2024, I propose that infinitives come in several sizes. In subject control, complements of typical restructuring predicates (modals, aspectual verbs, *try*) are argued to be either vP or AspP sized, whereas non-restructuring predicates are argued to take TP-sized complements. The two groups of infinitives line up closely with the well established distinction between restructuring/tenseless infinitives on the one hand, and non-restructuring/tensed infinitives on the other. All infinitival complements in Wolof, even those that are generally considered to be very small in some languages (e.g., VP-sized in German, according to Wurmbrand (2001)) can contain the morpheme *di*. In finite clauses *di* encodes imperfective aspect that carries present, habitual, or future meaning; Bochnak and Martinović (2018) treat it as an Asp head, and I assume that this is also the case in infinitives. Evidence for the fact that *di* involves a head from the inflectional layer lies in its interaction with clitic climbing, shedding further light onto properties of cliticization.

Clitics in Wolof obligatorily climb to the matrix C out of complements of typical restructuring verbs, such as modals, aspectual verbs, and implicatives, illustrated with a modal in (32).

(32) Clitic climbing with modals

a. War na=ñu [dimbale miskin yi].
must C=1PL.S [help poor the.PL]
'We must help the poor.'

b. War na=ñu=**leen** [dimbale].
must C=1PL.S=3PL.O [help]
'We must help them.'

c. *War na=ñu [dimbale=**leen**].
must C=1PL.S [help=3PL.O]

Clitic climbing is optional out of complements of verbs that cross-linguistically typically do not restructure, such as desideratives, as in (33); the object clitic *ko* can be found either after the embedded verb or attached to the matrix C, following the subject clitic (the optionality of clitic climbing is represented with curly brackets).

(33) Clitic climbing with desideratives

a. Taamu na=a [naan ataaya suba si].
prefer C=1SG.S [drink tea morning the.SG]
'I prefer to drink tea in the morning.'

b. Taamu na=a={**ko**} [naan={**ko**} suba si].
prefer C=1SG.S=3SG.O [drink=3SG.O morning the.SG]
'I prefer to drink it in the morning.'

In Martinović (2024) I propose that the optionality of clitic climbing is related to the presence of any head from the inflectional layer inside the infinitive—if such a head is absent, clitics obligatorily climb, and if it is present, they optionally climb. Complements of desiderative verbs are always (at least) TP-sized, therefore clitic climbing is always optional. This is in line with the claim that non-restructuring infinitives contain more structure than restructuring infinitives (e.g., Wurmbrand 2001 et. seq., Grano 2015). I argue that the presence of any head from the inflectional layer makes clitic climbing optional. Crucial evidence comes from the complements of otherwise obligatorily restructuring verbs, where the presence of the aspectual element *di* makes clitic climbing optional, just as in non-restructuring infinitives. (34a) shows this for clitic climbing out of the complement of the verb *war* 'must', and (34b) shows that the clitic can stay in the embedded infinitive which contains the verb *jéem* 'try', as long as *di* is present. I take this to mean that *di* is an inflectional head. Complements of typical restructuring verbs are therefore either vP-sized (when *di* is not present), and clitic climbing is then obligatory, or they are AspP-sized in the presence of *di*, when clitic climbing becomes optional.

(34) Clitic climbing always optional with *di*

a. War na=ñu={**ko**} [di={**ko**} dimbale].
must C=1PL.S=3SG.O [IPFV=3SG.O help]
'We must help him/her.'

b. War na=a={**ko**} [di={**ko**} jéem [togg]].
must C=1SG.S=3SG.O IPFV=3SG.O try cook
'I must try to cook it.'

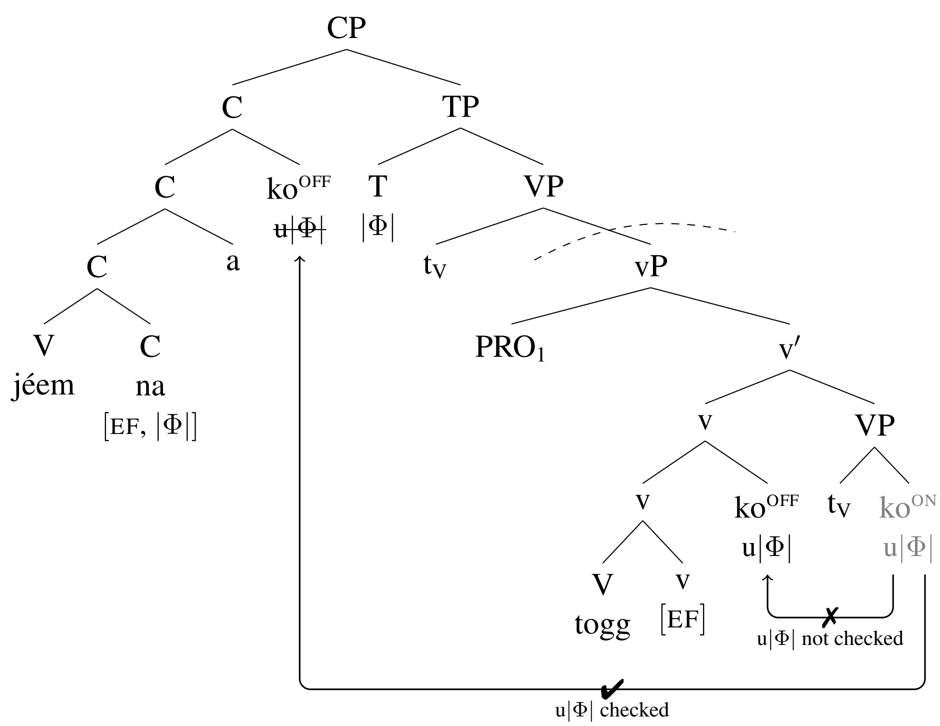
To capture the properties of clitic climbing, I propose that clitics have an additional feature that is checked by a head from the inflectional layer. The introduction of a feature specific to a particular layer of the clause follows the intuition that the clause is divided into 'domains', and that each domain is the locus of certain kinds of information/operations. One implementation of this observation is Grohmann's (2003) clausal tripartition into the *theta*-domain (roughly vP/VoiceP), *phi*-domain (roughly TP, or my 'inflectional layer'), and *omega*-domain, or the left periphery. Grohmann endows each head in a particular domain with a 'context value'. Following this, I propose that every head in the inflectional domain carries the feature $|\Phi|$, and that clitics have u$|\Phi|$. That clitics have a privileged relationship with the inflectional layer has been, in one way or another, a part of most analyses of clitic movement, though it is usually captured as a relationship with Agr or T (e.g., Kayne 1991), and often related to Case licensing. For example, Rezac (2005) proposes that clitics in Czech climb out of infinitives that are VP-sized (following Wurmbrand 2001), as there are no case licensers present in such a small structure, and clitics must be licensed by entering into a relationship with functional categories that they themselves lack, due to their structural deficiencies (along the lines of Cardinaletti and Starke 1999). All Wolof infinitives can have the aspect morpheme *di*, which means that no infinitive is only VP-sized, therefore an accusative case assigner would be present. Even in the presence of *di*, clitics can climb (though they do not have to). Additionally, it is not clear that v would be involved in the licensing of the locative clitic. Furthermore, if we take seriously the analyses of future-oriented infinitives, which are argued to be CP/TP-sized, or wollP-sized (e.g., Wurmbrand 2014a; Grano 2012, 2015), assuming that they must be reduced to VPs in the instances of clitic climbing should prevent future-oriented interpretations, if those are licensed by additional structure, as, for example, Wurmbrand argues (and proposes that this is what licenses mismatched adverbials in the matrix and embedded clause, e.g., *Frodo decided today to take the ring to Mordor tomorrow*). This is not what happens. To preserve the distinction argued to exist between restructuring infinitives, which do not allow future-oriented interpretation with respect to the matrix event (*Frodo tried today to leave the Shire* (*tomorrow)), and typically non-restructuring ones, that do, I keep the distinction in size between complements of typical restructuring verbs (smaller than TP), and complements of typical non-restructuring verbs (TP), and therefore argue that the relevant feature for clitics is a context feature $|\Phi|$ on every head in the inflectional layer, as opposed to, e.g., [φ] on C or T.

Specifically, I propose that any head from the inflectional layer can establish the required relationship with the clitic, and I assume that this is involved in the licensing of the clitic (I stay agnostic on the question of why such licensing is required). To illustrate, consider the following derivations of obligatory and optional clitic climbing from the infinitival complement of the restructuring verb *jéem* 'try' (from Martinović 2024, p.1599-1600). First, in the absence of aspect, clitics must climb to the matrix C. I propose that there are two EDGE-features, one on the highest head of the infinitival complement, here a vP, and one on the matrix C. Note that I do not claim that the EDGE-feature occurs on every v/phase head, only on the highest head in the extended projection of the verb. Given that I argue that complements of control verbs can be vP, AspP, or TP-sized, any of those heads can carry the EDGE-feature. If the clitic were to climb to the edge of the embedded infinitive, v, his would turn its switch off, but its u$|\Phi|$ feature would not be checked. Moving the clitic to the matrix C results in the checking of u$|\Phi|$, and I argue that this makes clitic climbing obligatory in restructuring constructions without *di*, illustrated in (35).

(35) Obligatory clitic climbing with *jéem 'try' without di*

a. Jéem na=a=**ko** [togg=(***ko**)].
try C=1SG.S=3SG.O cook=(*3SG.O)
'I tried to cook it.'

b.



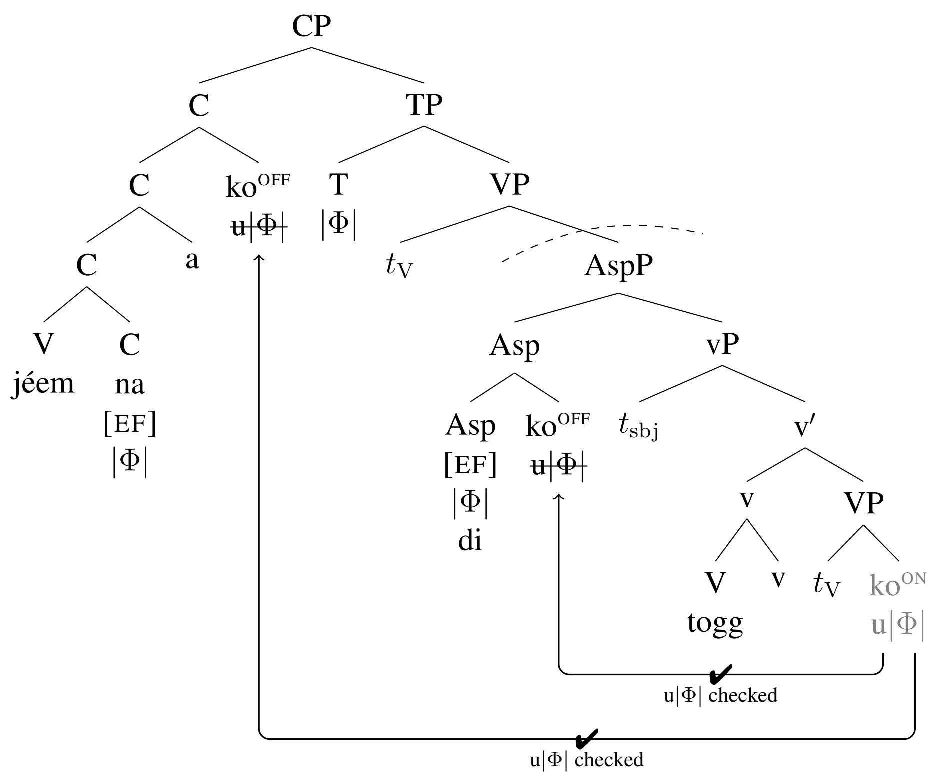
I have here represented the movement to the higher C as occurring in one fell swoop, seemingly incurring a Minimality violation, as it is skipping over the intervening [EF] on v. I return to the details of how movement to the higher clause proceeds in the following subsection.

In the presence of *di*, clitic climbing becomes optional, just as it is optional out of complements of non-restructuring verbs. In Martinović (2024), I take this as evidence for the similarity between the infinitival clause with *di* in (36), and the infinitival complement of a non-restructuring verb which I argue to be a full TP. Specifically, I propose that the context |Φ|-feature is found on any head in the inflectional layer, and therefore any head can check u|Φ| on the clitic. The clitic can therefore surface attached to the highest head in the infinitival complement, Asp, or it can attach to the matrix C.

(36) Optional clitic climbing with *jéem 'try' with di*

a. Jéem na=a={**ko**} [di={**ko**} togg].
try C=1SG.S=3SG.O IPFV=3SG.O cook
'I tried to cook it.'

b.



Clitic climbing is equally optional with typically non-restructuring subject control verbs, which I argue take a TP-sized complement, and would receive the same analysis as with the restructuring verb *jéem* 'try' in the presence of *di*. In the following subsection, I provide an analysis of the optionality of clitic climbing.

4.2 Optionality of clitic climbing

The optionality of clitic climbing introduces several problems, depending on whether clitic movement to the higher edge over a lower one happens in two steps, or in one fell swoop. If clitics first move to the lower edge and then to the higher one, and if they adjoin as heads, it means that they need to excorporate. In the following section, I argue that clitics must move as heads, as opposed to undergoing, for example, phrasal movement in the syntax followed by PF adjunction to the highest head. Additionally, if clitics can optionally stay at the lower edge, under the proposed analysis this indicates that their u$|\Phi|$ has been checked. The question then arises of how to even motivate their movement to the higher edge: if the EDGE-feature in the matrix clause turns the clitics' switch on again, we would expect their movement to be obligatory, not optional. And finally, if clitics can either stay at the edge of the infinitival clause or move to the matrix clause, we may expect that they could be split—some clitics might move to the higher edge, and some stay at the lower edge; splitting the clitic cluster has been reported to be possible in some varieties of Serbo-Croatian (Stjepanović 1998, 1999), but it is not possible in Wolof.

If, on the other hand, clitics skip the lower edge and move directly to the matrix C, they seem to be violating Relativized Minimality (Rizzi 1990). In languages that have optional clitic climbing (e.g., Italian), one account of the optionality has been that in the absence of clitic climbing we are dealing with a non-restructuring configuration (larger complement out of which clitics cannot climb), whereas in restructuring constructions the complement is very small (e.g., VP), so clitics obligatorily climb. In Wolof, clitic climbing is possible out of structures that are universally argued to be larger than VPs (complements of desiderative verbs), and climbing can occur in the presence of *di*, indicating that these complements cannot be as small as VPs.

I propose that clitics undergo only one instance of movement: either to the lower edge, or to the higher edge, and that there is no Minimality violation. I follow the theory of movement developed by Heck (2016, 2022, 2023), who argues that movement consists of two steps:

1. Removal of the element from the tree into a separate workspace (WSP, independently proposed in Bobaljik 1915; Uriagereka 1995; Müller 2017, 2018). Each workspace is associated with a specific feature.

2. Remerger of the element back into the tree.

If the two operations apply in succession, we cannot identify them; Heck explores a large number of phenomena in which another syntactic process intervenes between the removal and remerger, explaining why a particular structure does not violate either the Strict Cycle Condition (Chomsky 1973) or the Minimal Link Condition (Fanselow 1991; Ferguson 1993; Chomsky 1995), even though on the surface it appears to do just that. The crucial component of Heck's theory that makes this possible is the proposal that remerger can be delayed, or *procrastinated*,[10] so that removed elements can sit in the WSP while another operation occurs, and be remerged into the derivation later. Heck investigates cases where the head (or its specifier) that triggered the removal of an element into the WSP is also the position where the element ultimately remerges; I propose here that remerger in the case of cliticization can target a different position.

First, we investigate the option in which the clitic does not climb, as in (37). When the head with the EDGE-feature is merged, in this case T, it turns the clitic's switch ON. This triggers the removal of the clitic into a workspace associated with [EF]. In step 2, the clitic is remerged by being adjoined to T. The clitic's switch is turned OFF, and its u$|\Phi|$ is checked; I propose that this

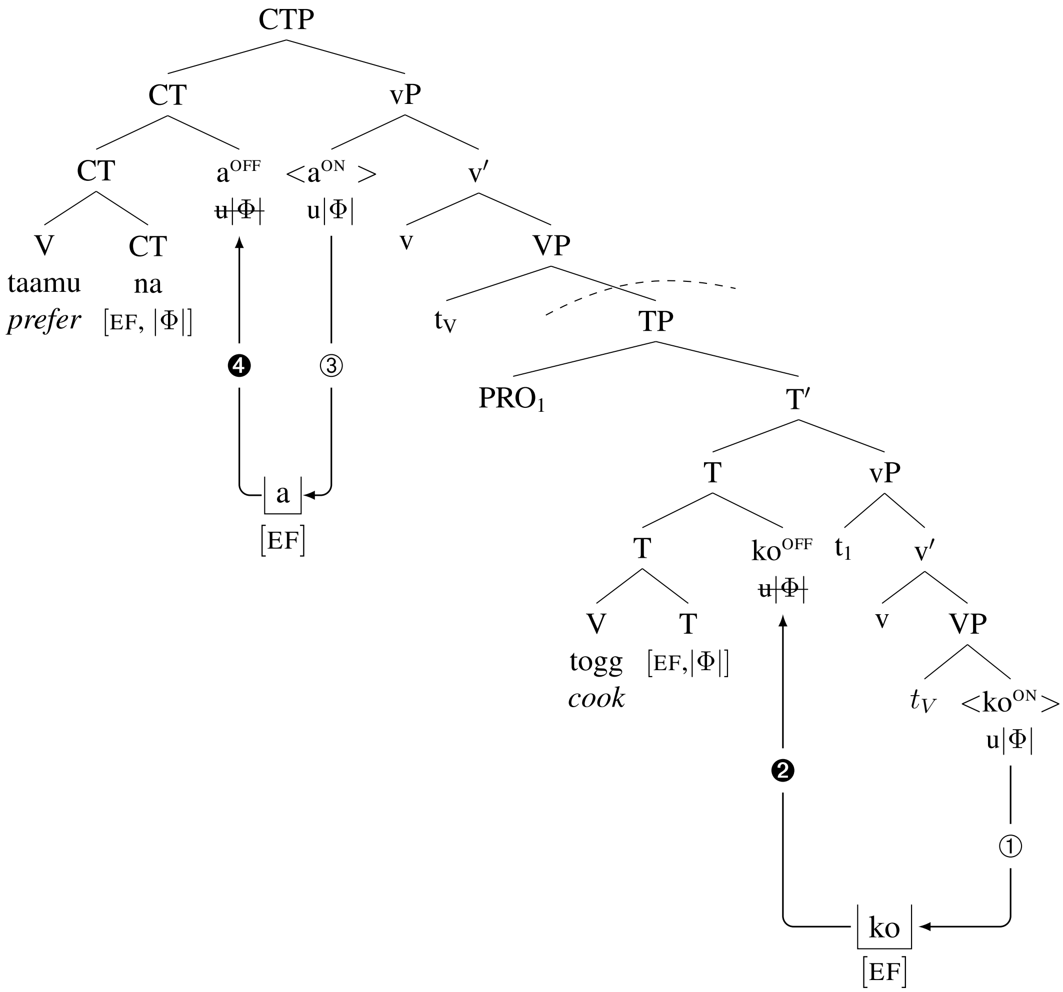
[10]Heck does not use the term in the sense of Chomsky (1993), who refers to movement that occurs at LF as being procrastinated. For Heck, procrastination is the delay of the second step of movement while an intervening operation takes place.

makes the clitic invisible to subsequent operations, so it cannot be attracted by the EDGE-feature on the matrix CT. The example in (37) also involves the cliticization of the matrix subject, which proceeds in the same manner.

(37) No clitic climbing

a. Taamu na=a [togg=ko].
prefer C=1SG.S cook=3SG.O
'I prefer to cook it.'

b.



The second option is for the object clitic, which was removed to the WSP by the embedded [EF] on T, not to remerge with the embedded T, but to procrastinate in the WSP and remerge with the higher head that is also associated with [EF].[11] I propose that remerger can procrastinate because the EDGE-feature is not a feature that needs to be checked, so not remerging the clitic in the embedded clause does not cause problems on that front; the only feature that does need to be checked is the u|Φ| on the clitic, and there is no requirement for it to be checked in the embedded clause. The EDGE-feature on the matrix C activates the switch on the matrix subject clitic (*a*),

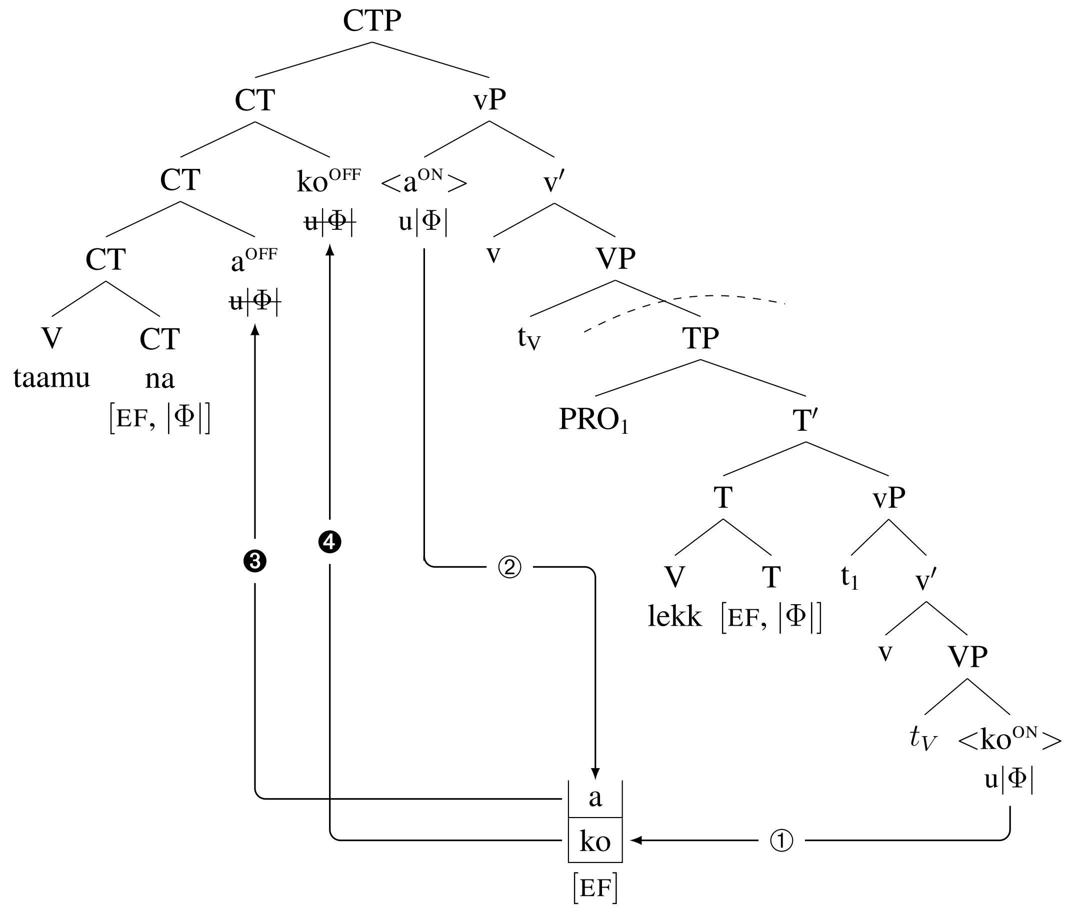
[11] This means we are dispensing with any sort of *Earliness Requirement for Feature Checking* requiring unsaturated features are saturated as early as possible; Heck's theory requires this to be the case.

which I assume is removed into the same workspace where *ko* has been procrastinating, as their removal is triggered by the same kind of feature. Finally, both clitics, the matrix subject and the embedded object, remerge with the matrix C.

(38) Clitic climbing

a. Taamu na=a=ko [togg].
prefer C=1SG.S=3SG.O cook
'I prefer to cook it.'

b.



Going back to obligatory clitic climbing, movement procrastination allows us to account for that case as well. In (35), there are two EDGE-features, one on the embedded v, and one on the matrix CT. I argued that clitic climbing is obligatory because the embedded v does not have |Φ|, so if the clitic were to remerge with v, its u|Φ| would remain unchecked. Given that I have argued that [EF] and u|Φ| independent, we have to assume that [EF] on v turns the clitic's switch ON in (35). If the clitic does not procrastinate but merges with v, its u|Φ| is not checked. If it procrastinates and merges with the matrix CT, both its u|Φ| is checked and its switch is turned OFF. The difference between obligatory and optional clitic climbing is that only the derivation with procrastination results in a grammatical output in obligatory clitic climbing, whereas either the derivation with or the one without procrastination is succesfull in optional clitic climbing.

In this section, we discussed another property of clitics, argued for in Martinović 2024—their need to occupy the inflectional layer of the clause. We also saw how cliticization can be a useful tool in diagnosing clause size. That infinitival complements of control verbs are of different sizes has been independently argued for in the literature on control (e.g., Wurmbrand 2001, 2014b, 2015; Grano 2015; Wurmbrand and Lohninger for.). The patterning of clitic climbing in Wolof lines up with the standard distinction between restructuring and non-restructuring verbs, and what is more, behaves as expected with changes of clause size. Finally, I have proposed a new analysis of the optionality of clitic climbing, which relies on Heck's (2016, 2022, 2023) theory of movement, which allows us to account for the apparent optionality of the movement in a principled manner, without violating Minimality or modifying head adjunction to allow excorporation of clitics (but not other heads in the language).

We now turn to infinitives with overt subjects, which further illuminate the properties of clitic movement, and especially the behavior of the subject clitic. I argue that this offers additional evidence against a prosodic component to clitic placement in Wolof.

5 Non-finite complements with overt subjects

In this section I offer what I believe to be a conclusive argument against a prosodic component to cliticization in Wolof, from infinitival complements with overt subjects. Such infinitives can be a complement clause, as in (39a), or an adjunct clause, as in (39b). The former structures do not exhibit obligatory control, whereas the latter ones do; this is not relevant here, but see Martinović 2024 for more details.

(39) Infinitives with overt subjects

a. Aaye na=a samay wayjur$_1$ [**ñu**$_{1/2}$ may xale yi tàngal].
prevent C=1SG.S POSS.1PL parents [3PL.S give child the.PL candy]
'I prevented my parents from giving candy to the children.'

b. Dimbali na=a sama yaay$_1$ [**mu**$_{1/*2}$ dem].
help C=1SG.S POSS.1SG mother 3SG.S leave
'I helped my mother to leave.'

Infinitives with overt subjects involve a CP-layer. There are two pieces of evidence for this. First, clitic climbing out of infinitives with overt subjects is ungrammatical, as shown in (40), and C is known to block clitic climbing (e.g., Wurmbrand 2001; Bondaruk 2004; Marušič 2005; Dotlačil 2007).[12]

(40) No clitic climbing from infinitives with subjects

a. Da=ma bëgg [Demba togg ceebujën].
C=1SG.S want Demba cook ceebujen
'I want Demba to cook ceebujen.'

b. Da=ma bëgg [Demba togg=**ko**].
C=1SG.S want Demba cook=3SG.O
'I want Demba to cook it.'

[12]Note that *Demba* in (40) is the embedded subject, and not the matrix object, as the pronoun that would surface in place of *Demba* is the subject 3rd person clitic *mu*, and not the object clitic *ko*.

c. *Da=ma=**ko** bëgg [Demba togg]
C=1SG.S=3SG.O want Demba cook

Second, also noted by Fong (2024), A′-extraction out of these kinds of infinitival complements results in resumption, as shown in (41) (data mine).

(41) Resumption in A′-movement out of infinitives with overt subjects
Lan la=∅ yey xarit$_1$-am [mu$_1$ togg=**ko**]?
what C=3SG.S convince friend-POSS.3SG 3SG.S cook=3SG.O
'What did s/he convince your friend to cook?'

A′-movement in Wolof requires the presence of the wh-complementizer *(l)a*, as in (42a) (Martinović 2015, 2017b), and resumption in such constructions, which pass all the standard movement tests, is impossible. An A′-dependency into a complement clause headed by a non-wh-C, for example *na*, requires the presence of a resumptive pronoun, as shown in (42b) (data from Torrence 2013), and in Martinović 2024 I argue that they do not involve A′-extraction.

(42) A′-movement requires the complementizer la
a. téeré$_1$ bi=leen=fa xale yi wax ne [$_{CP}$ $\boxed{\text{la}}$=a=(***ko$_1$**) jënd]
book C=3PL.O=LOC child the.PL say that C$_{WH}$=1SG.S=3SG.O buy
'the book that the children told them there that I bought'
b. téeré$_1$ bi=leen=fa xale yi wax ne [$_{CP}$ jënd $\boxed{\text{na}}$=ñu=***(ko$_1$)** démb]
book C=3PL.O=LOC child the.PL say that buy C=3PL.S=3SG.O yesterday
'the book that the children told them there that I bought it yesterday'

The presence of a resumptive pronoun in (41) indicates that A′-movement out of the infinitival clause is not possible. Since A′-movement is blocked by a non-wh-C, (41) suggests that a C head is present, and it is one without a wh-feature, just as *na* in (42b). Compare this to examples of subject control discussed in the previous section. As (43) shows, extraction out of an infinitival clause without an overt subject does not allow resumption, with or without clitic climbing, suggesting that those structures are never as big as CPs.

(43) No resumption in extraction out of subject control infinitives
a. Lan$_1$ la Demba taamu [jox=leen=(*ko$_1$)]?
what C$_{WH}$ Demba prefer give=3PL.O=3SG.O
'What did Demba prefer to give to them?'
b. Lan$_1$ la=leen(=*ko$_1$) Demba taamu [jox]?
what C$_{WH}$=3PL.O=3SG.O Demba prefer give
'What did Demba prefer to give to them?'

Infinitival clauses with overt subjects give additional support for the absence of a clearly identifiable prosodic requirement placed on clitics. First, since the subject clitic is initial in these infinitives, it cannot be that clitics must lean to a prosodic word on their left. Second, the lower clitics do not cluster with the subject clitic, as in (44d), but follow the highest head, in (44b). If there was a prosodic requirement on clitic placement, one would expect it to apply uniformly to all clitics.

(44) Clitics in infinitives with overt subjects

a. Da=ma=leen aaye [**ñu** may xale yi mango ci néeg bi].
C=1SG.S=3PL.O prevent [3PL.S give child the.PL mango in room the.SG]
'I prevented them from giving mango to the children in the room.'

b. Da=ma=leen aaye [**ñu** may=$\boxed{\textbf{fa}}$ xale yi mango].
C=1SG.S=3PL.O prevent [3PL.S give=LOC child the.PL mango]
'I prevented them from giving mango to the children there.'

c. *Da=ma=leen aaye [**ñu** may xale yi mango=$\boxed{\textbf{fa}}$].
C=1SG.S=3PL.O prevent [3PL.S give child the.PL mango=LOC]

d. *Da=ma=leen aaye [**ñu**=$\boxed{\textbf{fa}}$ may xale yi mango].
C=1SG.S=3PL.O prevent [3PL.S=LOC give child the.PL mango]

Note that the lower clitics in these structures do move. The PP *ci neeg bi* in (44) obligatorily follows the direct object *mango*, but the locative clitic must move to follow the verb, and cannot follow the object.

Any analysis that orders clitics in finite clauses to the right edge of C due to their prosodic status would need to make provisions for the subject clitic in infinitives. For example, we might say that it is only superficially like a clitic, but that it is actually a different kind of pronoun that does not undergo clitic movement. But then, if clitics were targeting the edge of a prosodic domain, we might expect that lower clitics should cluster with subjects. This is not the case, as shown in (44d), and (47c) below, with a clitic and a non-pronominal subject respectively. Even more problematically, a configuration in which all clitics cluster together exists, and that one seems to be the result of a prosodically motivated metathesis: when *di* occurs with the subject clitic, it becomes a glide, as in (45a), and if there is an object clitic, the glide metathesizes with the object, as in (45b).

(45) Infinitives with clitics and *di*

a. Da=ma=leen aaye [**ñu** **di** (>**ñuy**) jox xale yi tàngal].
C=1SG.S=3PL.O prevent 3PL.S IPFV give child the.PL candy
'I prevented them from giving candy to the children.'

b. Da=ma=leen aaye [**ñu=ko** **di** (>**ñukoy**) jox xale yi].
C=1SG.S=3PL.O prevent 3PL.S=3PL.O IPFV give child the.PL]
'I prevented them from giving it to the children.'

Note that the object clitic is not, in fact, moving to attach to the subject clitic. First, some speakers can leave the object clitic attached to *di*, as in (46a), though this is nobody's preferred form. Second, only clitics ending in vowels can cluster with the subject clitic, others must attach to *di*, shown in (46b)-(46c).

(46) a. %Da=ma=leen aaye [**ñu** di=$\boxed{\textbf{ko}}$ jox xale yi].
C=1SG.S=3PL.O prevent 3PL.S IPFV=3SG.O give child the.PL]
'I prevented them from giving it to the children.'

b. Da=ma=ko aaye [**ñu** di=$\boxed{\textbf{leen}}$ jox xale yi].
C=1SG.S=3SG.O prevent 3PL.S IPFV=3PL.O give child the.PL]
'I prevented her from giving them to the children.'

c. *Da=ma=ko aaye [**ñu=leen** di (>ñuleeni) jox xale yi].
C=1SG.S=3SG.O prevent 3PL.S=3PL.O IPFV give child the.PL]

Additionally, in infinitives with non-pronominal subjects and *di*, as in (47), the clitic obligatorily attaches to *di*, which in that case cannot become a glide.

(47) Infinitive with non-pronominal subject

a. Da=ma bëgg-ul [xale yi di lekk tàngal].
C=1SG.S want-NEG child the.PL IPFV eat candy
'I did not want the children to eat candy.'

b. Da=ma bëgg-ul [xale yi di=**ko** lekk].
C=1SG.S want-NEG child the.PL IPFV=3SG.O eat
'I did not want the children to eat it.'

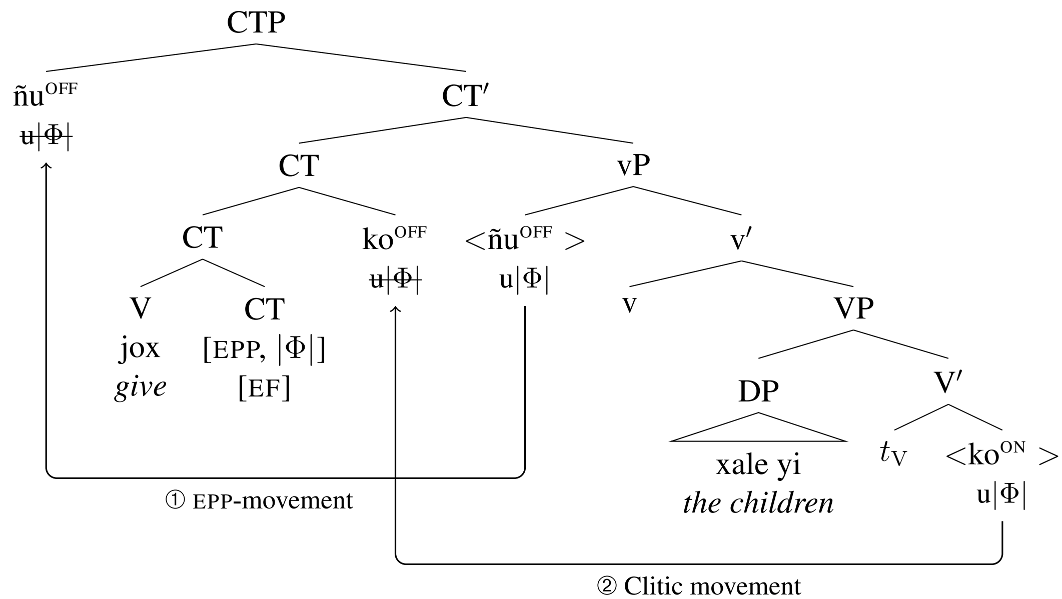
c. *Da=ma bëgg-ul [xale yi=**ko** di (>koy) lekk].
C=1SG.S want-NEG child the.PL=3SG.O IPFV eat

I believe these examples definitively exclude a prosodic component to clitic placement. First, we see that the subject clitic does not pattern with other clitics if it is located above the highest verbal element in the infinitive. We also see that other clitics can, as a result of cliticization of *di*, cluster with the subject clitic, meaning that, prosodically, the subject clitic is a suitable host. If clitics were placed at the left edge of the intonational phrase at PF, we would then expect them to always cluster with the subject clitic, since they are able to do so following phonological metathesis with the imperfective morpheme. What is more, there is no reason why they would not be positioned there in finite clauses as well, instead of following the sentence particle. Prosodically, clitics do not behave in a uniform manner. The only way to define the position that clitics target is in syntactic terms, as the highest head in the domain of Clitic Movement. This strongly favors a purely syntactic analysis of cliticization in Wolof. The remainder of this section is dedicated to spelling out the details of the behavior of the subject clitic, which can be explained if the subject can move as a phrase for EPP; no additional stipulations about the status or properties of subject clitics are needed under the present analysis.

In discussing A′-extraction, I have argued that the subject clitic must be able to move for EPP. Non-finite clauses with overt subjects give further evidence for this, as the subject is always overt in them, and if it is pronominal, it is a weak, not a strong pronoun. I also showed that infinitives that have overt subjects differ from subjectless infinitives: clitics cannot climb in the presence of an overt subject, and A′-movement is blocked from infinitives with overt subjects. I therefore argued that these infinitives contain the CP-layer. I propose that infinitives with overt subjects as in (48a) (and so-called *minimal clauses*, shown in (12)) have the structure in (48b). Just as in finite non-extraction clauses, I assume that C and T heads are bundled into one head. CT has an EPP-feature, and it is also the highest head in the extended projection of the verb, therefore also carrying the EDGE-feature. I propose that EPP triggers movement of the subject first, with the clitic moving to Spec,CTP to check it. The EDGE-feature operates last to turn the switch on the clitic ON. The subject clitic, however, is not in the c-command domain of [EF] anymore, so its switch remains OFF, it does not undergo Clitic Movement (i.e., adjunction to CT), and no requirement placed on the clitic is violated. The lower clitics move to attach to CT. Clitics also have their $|\Phi|$-feature checked—the object clitic by adjunction to CT, and the subject clitic via Spec-Head agreement—as these infinitives contain the inflectional layer.[13]

(48) a. Da=ma=leen aaye [**ñu** jox= **ko** xale yi].
C=1SG.S=3PL.O prevent 3PL.S give=3SG.O child the.PL
'I prevented them from giving it to the children.'

b.

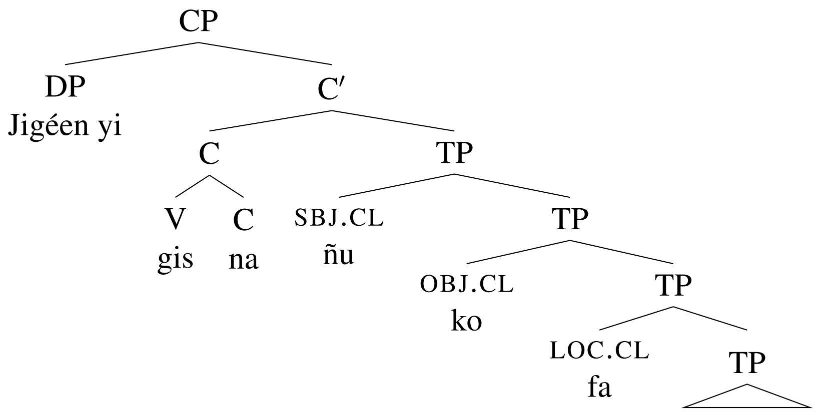


The behavior of subject clitics, which stay initial when there is no higher functional head for them to move to, but move like other clitics when there is, helps dispense with alternative analyses of cliticization which involve prosodic reordering of the kind that has been proposed for second position clitics in BCS (e.g., Halpern's (1995) Prosodic Inversion). In her thesis on Wolof cliticization, Russell (2006) proposes that the subject clitic in finite clauses first moves to Spec,TP, and that the object and locative clitics tuck in underneath the subject clitic, as in (49). The clitics are then displaced at PF as a cluster, so that they end up immediately following C; Russell does not motivate this PF displacement, so it is not clear what kind of a prosodic requirement the clitics are supposedly satisfying.

(49) a. Jigéen yi gis na=ñu=ko=fa.
woman the.PL see C=3PL.S=3SG.O=LOC
'The women saw her/him/it there.'

[13]Note that I assume that this CT head has an EPP-feature, whereas this was not the case for the CT head spelled out as *na* in finite clauses, given that its specifier may remain empty. The subject clitic therefore does not undergo EPP-movement in finite clauses, but adjoins to the sentence particle.

b.



While Russell's analysis can derive clitic distribution in finite clauses, it cannot account for clitic distribution in infinitives, as we would expect for all clitics to be either initial, or, if that violated some prosodic requirement placed on clitics, for all of them to be displaced at PF to follow the verb. Russell assumes that clitic movement to Spec,TP is triggered only by the finite T, and that clitics in infinitives do not move at all. We have seen clear evidence that this is not the case. Even if we do not subscribe to the analysis of infinitival clause size from Martinović 2024, infinitives with *di* show us that clitics do move: the subject must move to some higher position to precede *di*, and lower clitics must move over the lexical verb to follow *di*. Phrasal movement to some specifier, either a lower one as in Russell, or perhaps a higher one, would have to be followed by PF displacement, which would need to exempt the subject clitic in infinitival clauses, but not in finite ones. Given that we have no independent evidence that Wolof clitics are sensitive to prosody, this is a less desirable analysis. Movement of clitics to the highest head in the extended projection of the verb, coupled with EPP-triggered movement of the subject clitic, can fully account for the distribution of Wolof clitics.

If the analysis presented here is correct, then we must allow for (i) long head movement, and (ii) movement that adjoins to the right. I maintain that long head movement is necessary for cliticization, as has been argued elsewhere (e.g., Rezac 2008; Roberts 2010; Preminger 2019). As for right-adjoining movement, a reviewer suggests that it could be avoided if we concede a limited role of prosody and assume that all head adjunction operates in the same way. There is no linear order in Syntax and therefore no left vs. right adjunction, and linear order is fixed later at Spell-out. This is a possibility, however, numerous arguments for right-adjoining movement have been made in the literature, both head movement, which Pesetsky (2013) calls *Undermerge* (also Yuan 2017; Sun 2020), and complement-forming phrasal movement (Rosenbaum 1967; Postal 1974; McCloskey 1984; Kratzer 1995; Johnson 2000; Lin 2000; Sportiche 2005; Penka 2001). Unless all these cases can be reanalyzed as not involving right-adjunction, though, cliticization should not be singled out as the problematic case.

6 Conclusion

This paper explores the properties and placement of Wackernagel-like clitics in the Niger-Congo language Wolof. I have shown that Wolof clitics follow the highest head in the extended projection of the verb, as initially argued in Dunigan 1994, and that their placement can be accounted for in purely syntactic terms. Several arguments for the absence of any kind of a prosodic component in clitic placement have been given. First, clitics in finite clauses do not occupy a consistent position with respect to the left edge of the intonational phrase which contains them—they can either follow the first phonological word, or the second one. The second argument comes from the behavior of the subject clitic, which is initial in the extended projection of the verb if there is no higher functional head that could attract it, but patterns with other clitics when such a head is present. I have proposed that Clitic Movement is triggered by an EDGE-feature on the highest head in the extended projection of the verb, so that the feature on the clitics sensitive to the EDGE-feature becomes activated in its c-command domain, and stays inactive otherwise. This allows the subject clitic to be unaffected by Clitic Movement if it is in the specifier of the highest head, and speaks against any kind of a prosodic requirement that the clitics must satisfy. Finally, I show that all clitics can cluster at the beginning of the clause just in case they undergo a PF metathesis with the aspectual morpheme *di*, which becomes a glide in the context of the subject clitic. This drives home the point that clitics are not prosodically required to be enclitics, but can cluster at the very edge of the intonational phrase. An analysis where the clitics move to the highest head in the extended projection of the verb, and where the subject clitic can move to the specifier of that head for EPP, straightforwardly captures all the data.

I have additionally argued that clitics in Wolof must establish a structural relationship with an element in the inflectional layer of the clause based on patterns of clitic climbing, which is obligatory in restructuring constructions in which the non-finite complement does not contain any functional elements from the inflectional layer, but optional if it does. And finally, I have proposed a novel analysis of clitic climbing, relying on the theory of movement in Heck 2016, 2022, 2023, who proposes that it should be decomposed into two steps, the removal of an element into a separate workspace, and its subsequent remerger back into the derivation. The key element of Heck's theory, which allows him to account for various phenomena which seemingly violate the Minimal Link Condition and the Strict Cycle Condition, is that remerger can be procrastinated while an intervening operation takes place. I extend Heck's theory to cliticization, and propose that the remerger of clitics can target two different positions in clitic climbing configurations: the edge of the embedded infinitival clause, or the edge of the matrix clause. This allows us to capture the optionality of clitic climbing without having to contend with either a violation of Relativized Minimality, or with excorporation.

Wolof clitics confirm an important property of Wackernagel-like clitics: that their movement is primarily syntactic, as argued by Ouhalla (1989) for Berber and Franks (1998/2010) for Serbo-Croatian. And while final clitic placement in languages such as Serbo-Croatian does appear to involve prosodic readjustment, Clitic Movement of this type looks similar to other types of movement to the edges of domains. To the extent that such movements are syntactically motivated, so then is the movement of Wackernagel-like clitics.

Acknowledgements: I am grateful to the many Wolof speakers from St.-Louis, Senegal, who have shared their language with me over the years, especially Demba Lô, Mbaye Diop, Magatte N'Diaye, Jean-Léopold Diouf, Louis Camara, Lamine N'Diaye, Ahmet Fall, and Ousmane Guéye. I thank the audiences at the Departments of Linguistics at Cornell University and the University of British Columbia, where parts of this work were presented, Karlos Arregi for his feedback, and three anonymous reviewers for their thoughtful critique and suggestions. All errors are my own.

Competing Interests Statement: The author declares no competing interests.

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